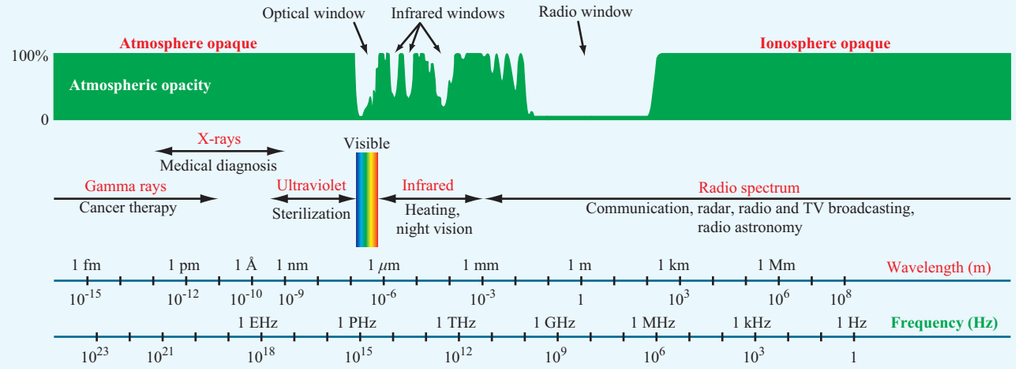


Figure 1-16 The electromagnetic spectrum.

the wavelength range covered by this band extends from 1 mm (300 GHz) to 1 cm (30 GHz).

Concept Question 1-9: What are the three fundamental properties of EM waves?

Concept Question 1-10: What is the range of frequencies covered by the microwave band?

Concept Question 1-11: What is the wavelength range of the visible spectrum? What are some of the applications of the infrared band?

1-6 Review of Complex Numbers

Any ***complex number*** z can be expressed in ***rectangular form*** as

$$z = x + jy, \tag{1.35}$$

where x and y are the ***real*** ($\Re\mathfrak{e}$) and ***imaginary*** ($\Im\mathfrak{m}$) parts of z, respectively, and $j = \sqrt{-1}$. That is,

$$x = \Re\mathfrak{e}(z), \qquad y = \Im\mathfrak{m}(z). \tag{1.36}$$

Alternatively, z may be cast in ***polar form*** as

$$z = |z|e^{j\theta} = |z|\angle\theta \tag{1.37}$$

where $|z|$ is the magnitude of z, θ is its phase angle,and $\angle\theta$ is a useful shorthand representation for $e^{j\theta}$. Applying ***Euler's identity***,

$$e^{j\theta} = \cos\theta + j\sin\theta, \tag{1.38}$$

we can convert z from polar form, as in Eq. (1.37), into rectangular form,

$$z = |z|e^{j\theta} = |z|\cos\theta + j|z|\sin\theta. \tag{1.39}$$

This leads to the relations

$$x = |z|\cos\theta, \qquad y = |z|\sin\theta, \tag{1.40}$$

$$|z| = \sqrt[+]{x^2 + y^2}, \qquad \theta = \tan^{-1}(y/x). \tag{1.41}$$

The two forms are illustrated graphically in **Fig. 1-18**. When using Eq. (1.41), care should be taken to ensure that θ is in the proper quadrant. Also note that, since $|z|$ is a positive quantity, only the positive root in Eq. (1.41) is applicable. This is denoted by the + sign above the square-root sign.

The ***complex conjugate*** of z, denoted with a star superscript (or asterisk), is obtained by replacing j (wherever it appears) with $-j$, so that

$$z^* = (x + jy)^* = x - jy = |z|e^{-j\theta} = |z|\angle -\theta. \tag{1.42}$$

The magnitude $|z|$ is equal to the positive square root of the product of z and its complex conjugate:

$$|z| = \sqrt[+]{zz^*}. \tag{1.43}$$

电磁波谱与1-6复数复习

图1-16见左侧原书：电磁波谱及大气透射窗口。下列中文图例与原图逐项对应。

图中：Gamma rays（伽马射线，癌症治疗）；X-rays（X射线，医学诊断）；Ultraviolet（紫外线，杀菌）；Visible（可见光）；Infrared（红外线，加热、夜视）；Radio spectrum（无线电波谱，通信、雷达、广播电视及射电天文）。Atmospheric opacity为大气不透明度；Optical、Infrared、Radio window分别为光学、红外、无线电窗口；Ionosphere opaque表示电离层不透明。下方刻度分别为波长（m）、频率（Hz）。

概念题1-9：电磁波具有哪三项基本性质？
概念题1-10：原书采用的微波频率范围是什么？
概念题1-11：可见光波长范围是多少？红外波段有哪些应用？

1-6 复数复习

任意复数可写为直角坐标形式z = x + jy；x、y分别是实部和虚部，$j^2 = -1$：

$$z = x + jy, \tag{1.35}$$

$$x = \Re\mathfrak{e}(z), \qquad y = \Im\mathfrak{m}(z). \tag{1.36}$$

也可写为极坐标形式；|z|为模，θ为辐角，∠θ为$e^{j\theta}$的简写。利用欧拉公式，可在两种形式之间转换：

$$z = |z|e^{j\theta} = |z|\angle\theta \tag{1.37}$$

$$e^{j\theta} = \cos\theta + j\sin\theta, \tag{1.38}$$

$$z = |z|e^{j\theta} = |z|\cos\theta + j|z|\sin\theta. \tag{1.39}$$

由此得到直角坐标与极坐标之间的关系：

$$x = |z|\cos\theta, \qquad y = |z|\sin\theta, \tag{1.40}$$

$$|z| = \sqrt[+]{x^2 + y^2}, \qquad \theta = \tan^{-1}(y/x). \tag{1.41}$$

如图1-18，求θ时须根据x、y的符号确定象限，不能只取arctan(y/x)的主值。|z|为非负实数，应取平方根的非负值。

复共轭z*是在z的直角坐标表达式中把j换成−j；其模不变，辐角变号：

$$z^* = (x + jy)^* = x - jy = |z|e^{-j\theta} = |z|\angle -\theta. \tag{1.42}$$

复数的模还可由z与其共轭的乘积求得：

$$|z| = \sqrt[+]{zz^*}. \tag{1.43}$$

数学校注：z = 0时|z| = 0，辐角无定义；非零复数的辐角以2π为周期。计算可用atan2(y,x)正确处理象限。

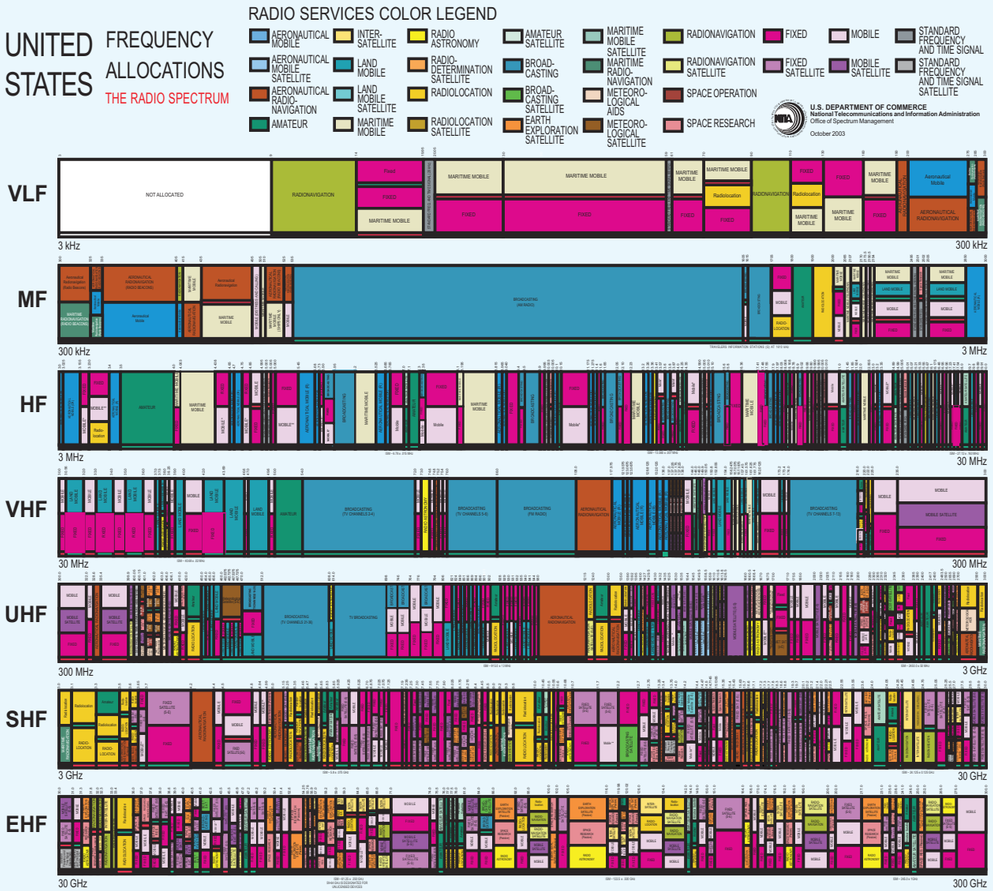


Figure 1-17 Individual bands of the radio spectrum and their primary allocations in the U.S. [See expandable version on book website: em8e.eecs.umich.edu.]

图1-17：无线电频谱图的中文对照

左侧保留原书完整频谱分配图，其来源为美国商务部国家电信和信息管理局（NTIA），日期为2003年10月。它说明当时美国无线电频谱的业务划分；原书提示可在配套网站查看放大版。

常用频段名称（对应原书命名）

缩写	中文名称	频率范围
VLF	甚低频	3—30 kHz
LF	低频	30—300 kHz
MF	中频	300 kHz—3 MHz
HF	高频	3—30 MHz
VHF	甚高频	30—300 MHz
UHF	特高频	300 MHz—3 GHz
SHF	超高频	3—30 GHz
EHF	极高频	30—300 GHz

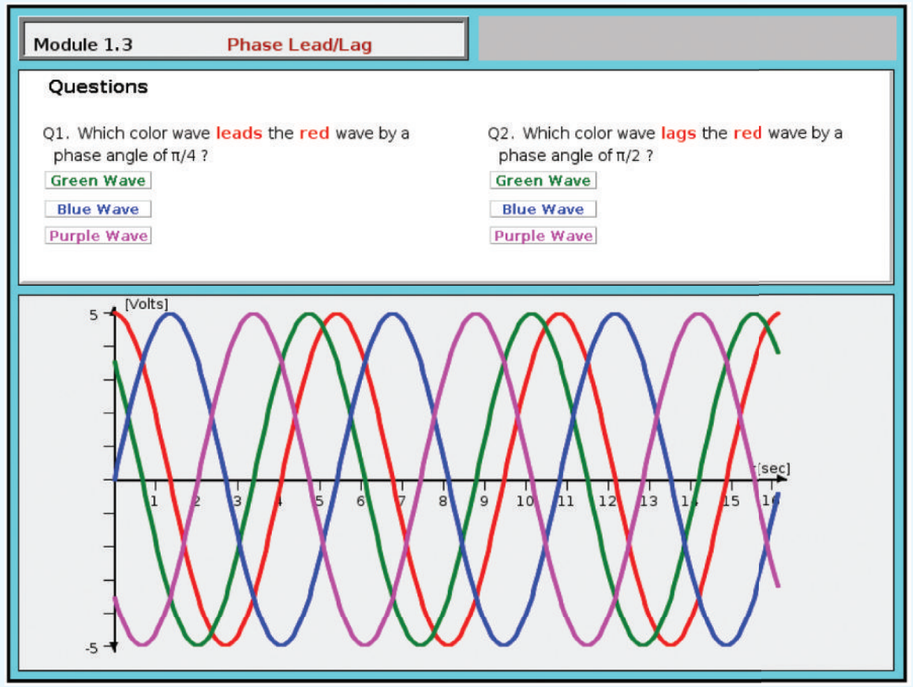
图中业务与标注词汇

英文	中文	英文	中文
Aeronautical mobile	航空移动	Aeronautical radionavigation	航空无线电导航
Amateur	业余	Broadcasting	广播
Earth exploration satellite	地球探测卫星	Fixed	固定
Inter-satellite	卫星间	Land mobile	陆地移动
Maritime mobile	水上移动	Maritime radionavigation	水上无线电导航
Meteorological aids	气象辅助	Meteorological satellite	气象卫星
Mobile	移动	Radio astronomy	射电天文
Radiodetermination satellite	无线电测定卫星	Radiolocation	无线电定位
Radionavigation	无线电导航	Space operation	空间操作
Space research	空间研究	Standard frequency and time signal	标准频率和时间信号
Satellite	卫星	Not allocated	未划分
Distress and calling	遇险与呼叫	Radio beacons	无线电信标
Telemetry / telemetering	遥测	Deep space	深空
Active / passive	有源 / 无源	E-S / S-E / S-S	地对空 / 空对地 / 空对空

复合业务名称按词组合理解，例如Fixed satellite为固定卫星业务，Broadcasting satellite为广播卫星业务，Aeronautical mobile satellite为航空移动卫星业务。图中的频率数字及业务分配位置保留在左侧原图，避免重绘误改。

年代校注：该图是2003年的美国频谱划分示意，不是现行规定，也不能作为其他国家的频率使用许可依据。本页补充完整图例词汇，未将历史分配表改写为当前分配表。

Module 1.3 Phase Lead/Lag Examine sinusoidal waveforms with different values of the reference phase constant ϕ_0.



We now highlight some of the properties of complex algebra that will be encountered in future chapters.

Equality: If two complex numbers z_1 and z_2 are given by

$$z_1 = x_1 + jy_1 = |z_1|e^{j\theta_1}, \tag{1.44}$$

$$z_2 = x_2 + jy_2 = |z_2|e^{j\theta_2}, \tag{1.45}$$

then $z_1 = z_2$ if and only if $x_1 = x_2$ and $y_1 = y_2$ or, equivalently, $|z_1| = |z_2|$ and $\theta_1 = \theta_2$.

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2). \tag{1.46}$$

Multiplication:

$$z_1 z_2 = (x_1 + jy_1)(x_2 + jy_2)$$
$$= (x_1x_2 - y_1y_2) + j(x_1y_2 + x_2y_1), \tag{1.47a}$$

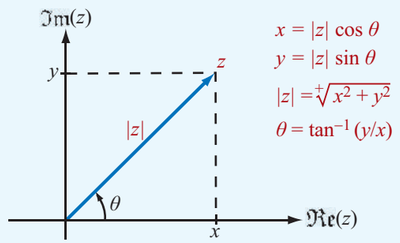
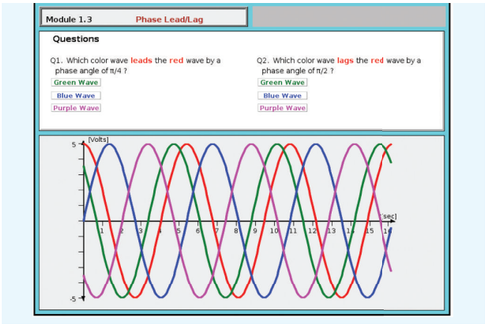


Figure 1-18 Relation between rectangular and polar representations of a complex number $z = x + jy = |z|e^{j\theta}$.

模块1.3与复数代数

模块1.3 相位超前 / 滞后：观察参考相位常数φ_0不同时的正弦波形。



原书模块1.3：相位差对时间波形的影响。

下面列出后续章节将使用的复数运算性质。设：

$$z_1 = x_1 + jy_1 = |z_1|e^{j\theta_1}, \tag{1.44}$$

$$z_2 = x_2 + jy_2 = |z_2|e^{j\theta_2}, \tag{1.45}$$

相等：$z_1 = z_2$当且仅当实部相等、虚部也相等。对于非零复数，等价条件是模相等，辐角在模2π意义下相等。

加法：实部与实部相加，虚部与虚部相加。

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2). \tag{1.46}$$

乘法：按代数展开，并利用$j^2 = -1$。

$$z_1 z_2 = (x_1 + jy_1)(x_2 + jy_2)$$
$$= (x_1x_2 - y_1y_2) + j(x_1y_2 + x_2y_1), \tag{1.47a}$$

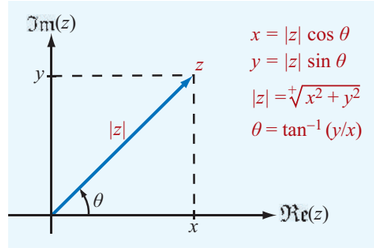


图1-18 复数z的直角坐标与极坐标表示：x为实部，y为虚部，|z|为模，θ为辐角。

原书“$\theta_1 = \theta_2$”的说法采用了相同辐角分支的约定；相差2nπ的辐角表示同一个非零复数。零复数的辐角没有定义。

or

$$\begin{aligned} z_1 z_2 &= |z_1| e^{j\theta_1} \cdot |z_2| e^{j\theta_2} \\ &= |z_1||z_2| e^{j(\theta_1+\theta_2)} \\ &= |z_1||z_2|[\cos(\theta_1+\theta_2) + j\sin(\theta_1+\theta_2)]. \end{aligned} \tag{1.47b}$$

Division: For $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{x_1+jy_1}{x_2+jy_2} = \frac{(x_1+jy_1)}{(x_2+jy_2)} \cdot \frac{(x_2-jy_2)}{(x_2-jy_2)} = \frac{(x_1x_2+y_1y_2)+j(x_2y_1-x_1y_2)}{x_2^2+y_2^2}, \tag{1.48a}$$

or

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{|z_1|e^{j\theta_1}}{|z_2|e^{j\theta_2}} = \frac{|z_1|}{|z_2|} e^{j(\theta_1-\theta_2)} \\ &= \frac{|z_1|}{|z_2|}[\cos(\theta_1-\theta_2) + j\sin(\theta_1-\theta_2)]. \end{aligned} \tag{1.48b}$$

Powers: For any positive integer n,

$$\begin{aligned} z^n &= (|z|e^{j\theta})^n \\ &= |z|^n e^{jn\theta} = |z|^n(\cos n\theta + j\sin n\theta), \end{aligned} \tag{1.49}$$

$$z^{1/2} = \pm|z|^{1/2} e^{j\theta/2} = \pm|z|^{1/2}[\cos(\theta/2) + j\sin(\theta/2)]. \tag{1.50}$$

Useful Relations:

$$-1 = e^{j\pi} = e^{-j\pi} = 1\angle 180^\circ, \tag{1.51}$$

$$j = e^{j\pi/2} = 1\angle 90^\circ, \tag{1.52}$$

$$-j = -e^{j\pi/2} = e^{-j\pi/2} = 1\angle -90^\circ, \tag{1.53}$$

$$\sqrt{j} = (e^{j\pi/2})^{1/2} = \pm e^{j\pi/4} = \frac{\pm(1+j)}{\sqrt{2}}, \tag{1.54}$$

$$\sqrt{-j} = \pm e^{-j\pi/4} = \frac{\pm(1-j)}{\sqrt{2}}.$$

Example 1-3: Working with Complex Numbers

Given two complex numbers

$$V = 3 - j4, \qquad I = -(2+j3),$$

(a) express V and I in polar form, and find (b) VI, (c) VI^*, (d) V/I, and (e) $\sqrt{I}$.

Solution:

(a) $$|V| = \sqrt[+]{VV^*} = \sqrt[+]{(3-j4)(3+j4)} = \sqrt[+]{9+16} = 5,$$

$$\theta_V = \tan^{-1}(-4/3) = -53.1^\circ,$$

$$V = |V|e^{j\theta_V} = 5e^{-j53.1^\circ} = 5\angle -53.1^\circ,$$

$$|I| = \sqrt[+]{2^2+3^2} = \sqrt[+]{13} = 3.61.$$

Since $I = (-2-j3)$ is in the third quadrant in the complex plane (Fig. 1-19),

$$\theta_I = 180^\circ + \tan^{-1}\left(\tfrac{3}{2}\right) = 236.3^\circ,$$

$$I = 3.61\angle 236.3^\circ.$$

(b) $$VI = 5e^{-j53.1^\circ} \times 3.61e^{j236.3^\circ} = 18.03e^{j(236.3^\circ - 53.1^\circ)} = 18.03e^{j183.2^\circ}.$$

(c) $$VI^* = 5e^{-j53.1^\circ} \times 3.61e^{-j236.3^\circ} = 18.03e^{-j289.4^\circ} = 18.03e^{j70.6^\circ}.$$

(d) $$\frac{V}{I} = \frac{5e^{-j53.1^\circ}}{3.61e^{j236.3^\circ}} = 1.39e^{-j289.4^\circ} = 1.39e^{j70.6^\circ}.$$

(e) $$\sqrt{I} = \sqrt{3.61e^{j236.3^\circ}} = \pm\sqrt{3.61}\, e^{j236.3^\circ/2} = \pm 1.90e^{j118.15^\circ}.$$

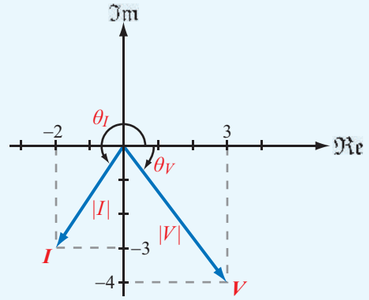


Figure 1-19 Complex numbers V and I in the complex plane (Example 1-3).

或

$$\begin{aligned} z_1 z_2 &= |z_1| e^{j\theta_1} \cdot |z_2| e^{j\theta_2} \\ &= |z_1||z_2| e^{j(\theta_1+\theta_2)} \\ &= |z_1||z_2|[\cos(\theta_1+\theta_2) + j\sin(\theta_1+\theta_2)]. \end{aligned} \tag{1.47b}$$

除法：设$z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{x_1+jy_1}{x_2+jy_2} = \frac{(x_1+jy_1)}{(x_2+jy_2)} \cdot \frac{(x_2-jy_2)}{(x_2-jy_2)} = \frac{(x_1x_2+y_1y_2)+j(x_2y_1-x_1y_2)}{x_2^2+y_2^2}, \tag{1.48a}$$

或

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{|z_1|e^{j\theta_1}}{|z_2|e^{j\theta_2}} = \frac{|z_1|}{|z_2|} e^{j(\theta_1-\theta_2)} \\ &= \frac{|z_1|}{|z_2|}[\cos(\theta_1-\theta_2) + j\sin(\theta_1-\theta_2)]. \end{aligned} \tag{1.48b}$$

幂运算：n为正整数，

$$\begin{aligned} z^n &= (|z|e^{j\theta})^n \\ &= |z|^n e^{jn\theta} = |z|^n(\cos n\theta + j\sin n\theta), \end{aligned} \tag{1.49}$$

$$z^{1/2} = \pm|z|^{1/2} e^{j\theta/2} = \pm|z|^{1/2}[\cos(\theta/2) + j\sin(\theta/2)]. \tag{1.50}$$

常用关系式：

$$-1 = e^{j\pi} = e^{-j\pi} = 1\angle 180^\circ, \tag{1.51}$$

$$j = e^{j\pi/2} = 1\angle 90^\circ, \tag{1.52}$$

$$-j = -e^{j\pi/2} = e^{-j\pi/2} = 1\angle -90^\circ, \tag{1.53}$$

$$\sqrt{j} = (e^{j\pi/2})^{1/2} = \pm e^{j\pi/4} = \frac{\pm(1+j)}{\sqrt{2}}, \tag{1.54}$$

$$\sqrt{-j} = \pm e^{-j\pi/4} = \frac{\pm(1-j)}{\sqrt{2}}.$$

例1-3 复数运算

已知两个复数：

$$V = 3 - j4, \qquad I = -(2+j3),$$

（a）将V、I写成极坐标形式；求（b）VI、（c）VI*、（d）V/I、（e）I的两个平方根。

解：

(a) $$|V| = \sqrt[+]{VV^*} = \sqrt[+]{(3-j4)(3+j4)} = \sqrt[+]{9+16} = 5,$$

$$\theta_V = \tan^{-1}(-4/3) = -53.1^\circ,$$

$$V = |V|e^{j\theta_V} = 5e^{-j53.1^\circ} = 5\angle -53.1^\circ,$$

$$|I| = \sqrt[+]{2^2+3^2} = \sqrt[+]{13} = 3.61.$$

I = −2 − j3位于复平面第三象限（图1-19），因此

$$\theta_I = 180^\circ + \tan^{-1}\left(\tfrac{3}{2}\right) = 236.3^\circ,$$

$$I = 3.61\angle 236.3^\circ.$$

(b) $$VI = 5e^{-j53.1^\circ} \times 3.61e^{j236.3^\circ} = 18.03e^{j(236.3^\circ - 53.1^\circ)} = 18.03e^{j183.2^\circ}.$$

(c) $$VI^* = 5e^{-j53.1^\circ} \times 3.61e^{-j236.3^\circ} = 18.03e^{-j289.4^\circ} = 18.03e^{j70.6^\circ}.$$

(d) $$\frac{V}{I} = \frac{5e^{-j53.1^\circ}}{3.61e^{j236.3^\circ}} = 1.39e^{-j289.4^\circ} = 1.39e^{j70.6^\circ}.$$

(e) $$\sqrt{I} = \sqrt{3.61e^{j236.3^\circ}} = \pm\sqrt{3.61}\, e^{j236.3^\circ/2} = \pm 1.90e^{j118.15^\circ}.$$

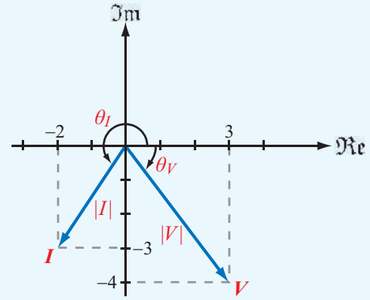


图1-19 复数V和I在复平面中的位置（例1-3）。

校对：VI = −18 − j；VI* = 6 + j17；V/I = (6 + j17)/13。平方根须保留±两支，不能把变量I翻译成汉字。

Technology Brief 2: Solar Cells

A ***solar cell*** is a photovoltaic device that converts solar energy into electricity. The conversion process relies on the ***photovoltaic effect***, which was first reported by 19-year-old Edmund Bequerel in 1839 when he observed that a platinum electrode produced a small current if exposed to light. The photovoltaic effect is often confused with the ***photoelectric effect***; they are interrelated, but not identical (**Fig. TF2-1**).

The photoelectric effect explains the mechanism responsible for why an electron is ejected by a material in consequence to a photon incident upon its surface (**Fig. TF2-1**(a)). For this to happen, the photon energy E (which is governed by its wavelength through $E = hc/\lambda$, with h being Planck's constant and c the velocity of light) has to exceed the binding energy with which the electron is held by the material. For his 1905 quantum-mechanical model of the photoelectric effect, Albert Einstein was awarded the 1921 Nobel Prize in physics.

Whereas a single material is sufficient for the photoelectric effect to occur, at least two adjoining materials with different electronic properties (to form a junction that can support a voltage across it) are needed to establish a ***photovoltaic current*** through an external load (**Fig. TF2-1**(b)). Thus, the two effects are governed by the same quantum-mechanical rules associated with how photon energy can be used to liberate electrons away from their hosts, but the followup step of what happens to the liberated electrons is different in the two cases.

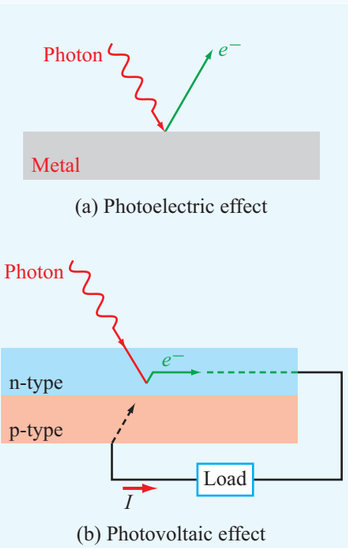


Figure TF2-1 Comparison of photoelectric effect with the photovoltaic effect.

The PV Cell

Today's photovoltaic (PV) cells are made of semiconductor materials. The basic structure of a PV cell consists of a ***p-n junction*** connected to a load (**Fig. TF2-2**).

Typically, the n-type layer is made of silicon doped with a material that creates an abundance of negatively charged atoms, and the p-type layer also is made of silicon but doped with a different material that creates an abundance of holes (atoms with missing electrons). The combination of the two layers induces an electric field across the junction, so when an incident photon liberates an electron, the electron is swept under the influence of the electric field through the n-layer and out to the external circuit connected to the load.

The ***conversion efficiency*** of a PV cell depends on several factors, including the fraction of the incident light that gets absorbed by the semiconductor material, as opposed to getting reflected by the n-type front surface or transmitted through to the back conducting electrode. To minimize the reflected component, an antireflective coating usually is inserted between the upper glass cover and the n-type layer (**Fig. TF2-2**).

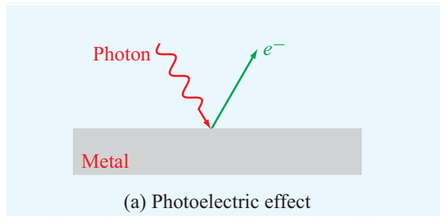
The PV cell shown in **Fig. TF2-2** is called a ***single-junction cell*** because it contains only one p-n junction. The semiconductor material is characterized by a quantity called its ***band gap energy***, which is the amount of energy needed to free an electron away from its host atom. For that to occur, the wavelength of the incident photon (which in turn defines its energy) has to be such that the photon's energy exceeds the band gap of the material. Solar energy extends over a broad spectrum, so only a fraction of the solar spectrum (photons with energies greater than the band gap) is absorbed by a single-junction material. To overcome this limitation, multiple p-n layers can be cascaded together to form a ***multijunction PV device*** (**Fig. TF2-3**). The cells usually are arranged such that the top cell has the highest band gap energy, thereby capturing the high-energy (short-wavelength) photons, followed by the cell with the next lower band gap, and so on.

技术短文2：太阳能电池

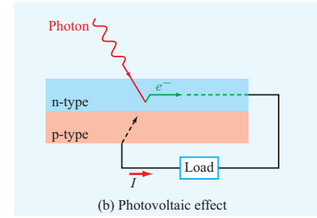
太阳能电池是一种把太阳能转换为电能的光伏器件。其转换过程利用光伏效应：1839年，19岁的贝克勒尔观察到铂电极受光照后产生微小电流，并报告了这一现象。光伏效应与光电效应有关，但二者并不相同（图TF2-1）。

本段所说的**光电效应**是外光电效应：入射光子使电子从材料表面逸出。光子能量E = hc/λ，其中h为普朗克常数，c为真空光速，λ为真空波长。电子逸出要求光子能量达到材料的逸出功。爱因斯坦于1905年提出光电效应的光量子解释，并因对光电效应定律的发现获1921年诺贝尔物理学奖。

单一材料即可发生外光电效应。原书用两种电子特性不同、相互接触形成结的区域说明光伏效应：结两端能建立电压，接入负载后便可产生光生电流。两种效应都涉及光子能量对电子的激发，但激发后电子的运动过程不同。



图TF2-1(a) 外光电效应。Photon：光子；Metal：金属；e⁻：电子。



图TF2-1(b) 光伏效应。n-type、p-type：n型、p型；Load：负载；I：电流。

光伏电池（PV cell）

光伏电池使用半导体材料，典型结构是与外部负载相连的p–n结（图TF2-2）。在硅中掺入施主杂质可形成以电子为多数载流子的n型区；掺入受主杂质可形成以空穴为多数载流子的p型区。两区接触后，结区形成内建电场。光吸收产生的电子和空穴被分离、收集，电子可经n型区的电极进入外部负载回路。

转换效率受多种因素影响，其中之一是入射光中被半导体吸收的比例，其余光可能在前表面反射，或透射至背电极。为减小反射，通常在玻璃盖板与n型层之间设置减反射膜（图TF2-2）。

图TF2-2只有一个p–n结，称为单结电池。半导体的带隙能量决定其带间光吸收的能量阈值。太阳光覆盖较宽光谱，低于带隙的光子不能通过通常的带间吸收产生电子–空穴对，因此单结电池不能利用整个太阳光谱。多结器件把不同带隙的子电池层叠组合（图TF2-3）：最高带隙的顶层吸收高能、短波光子，随后各层以逐渐降低的带隙利用透过的较低能光子。

学术校注：原文把n型掺杂说成产生“大量带负电的原子”，并把空穴称为“缺少电子的原子”，均不准确。n型多数载流子是电子，空穴是价带中的电子空态；电离施主带正电，均匀半导体体区通常近似电中性。带隙是导带底与价带顶的能量差，不是孤立原子的电离能。“两种材料”也不必是两种不同化学物质，同种硅的不同掺杂区即可形成p–n结。以上正文已按这些区别订正，依据链接见校对进度文件。